URBIEE 2026 — URBAN INNOVATION AND ENTREPRENEUR EXHIBITION — BRIDGING IDEAS, BUILDING FUTURES

GINKGO — FLOOD COMMAND CENTER

UNIVERSITI TEKNOLOGI MARA

GINKGO

FLOOD COMMAND CENTER: FROM RISK MAP TO EVACUATION PLAN FOR SELANGOR

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ABSTRACT

When floods come, people hear that a warning is in force, but not when to leave or where to go, and nobody can see which shelters will fill up. Ginkgo is a web-based flood command center for Selangor's nine districts that closes this gap.

Planning mode maps where river flooding is most likely, from satellite terrain, rivers, 38 years of surface-water records and census data on a grid of about 92 m cells. Flood drill mode then replays a storm hour by hour: it warns each neighbourhood on Japan's five-level scale, finds a walking or driving route along roads that stay dry to a centre that still has space, and says when to leave. A 3D model of Taman Sri Muda shows the water on the street, and an AI assistant answers in English or Malay with every figure checked against the data.

It runs in a browser on open data (Copernicus, OpenStreetMap, JRC, ESA, Meta, DOSM), so no GIS software is needed. The drill is a simulation for rehearsal, not a forecast.

OBJECTIVE

- Map where river flooding is most likely in Selangor under Normal, Heavy and Extreme rain, in a form anyone can read
- Rehearse a flood hour by hour, with alerts on Japan's five-level scale and a one-hour look-ahead for each neighbourhood
- Guide each household along a dry route to an evacuation centre that still has space, and say when to leave
- Show the water on the street in 3D, and let anyone ask in English or Malay with every number checked against the data

24.1%
of Selangor's land has High flood susceptibility

~1.93M
people live on land at High risk (estimate)

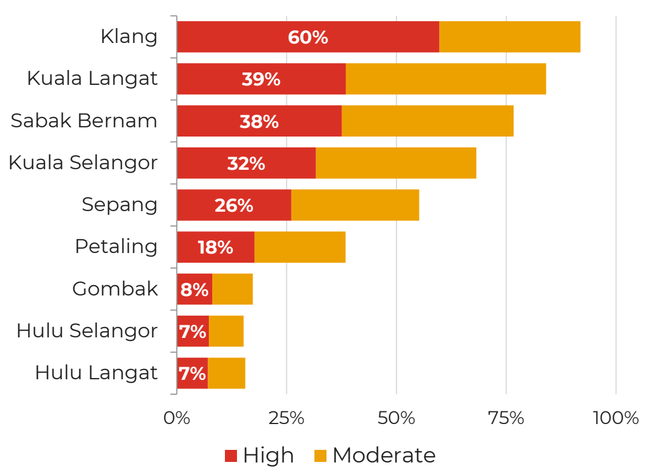
1,274
candidate evacuation centres, each with a capacity

USEFULNESS & COMMERCIALISATION POTENTIAL

- Turns open satellite and census data into a risk map, a drill and routes that anyone can use
- Gives each household a dry route, a centre with space and a leave-by time
- Open data and a browser app keep costs low: an AI answer is a few US cents, repeats are free
- Next: a council pilot on live river gauges, then Malaysia and ASEAN

RISK BY DISTRICT

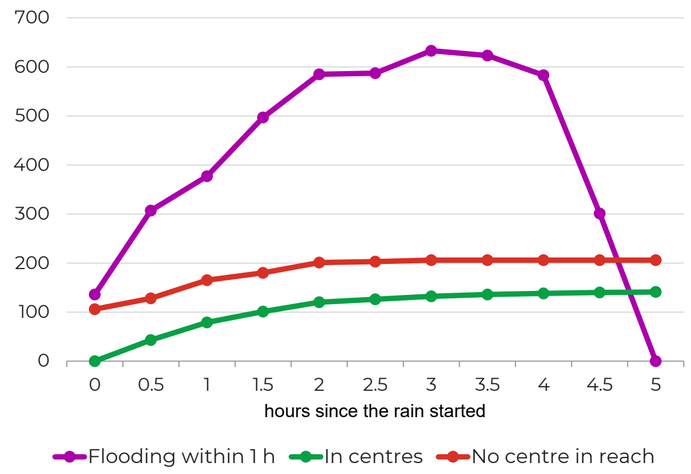
High and Moderate share of land



Source: Ginkgo analysis of open data

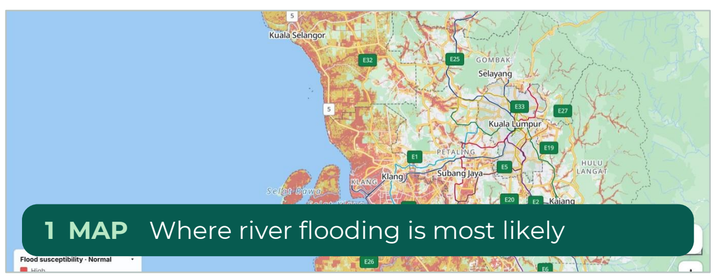
THE EVACUATION GAP

Klang storm drill, people (thousands)

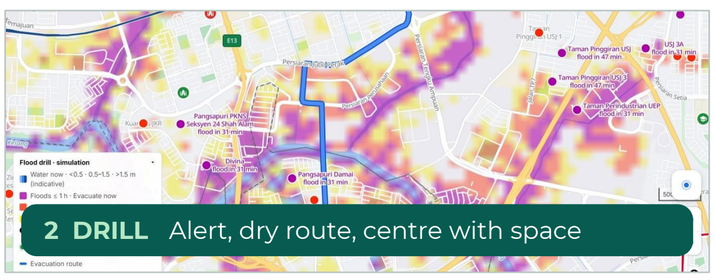


Source: Ginkgo flood drill, Extreme rain (estimate)

SYSTEM IN ACTION



1 MAP Where river flooding is most likely



2 DRILL Alert, dry route, centre with space



3 3D The water on the street

CONCLUSION

Ginkgo turns a flood map into a plan: when each neighbourhood must leave, where there is space, and how to get there on dry roads. Checked against 7 reported December 2021 flood sites (4 flagged High, 2 of 2 hilly controls Low). A simulation for rehearsal, not a forecast.